Entanglement Equilibrium and 2D Dilaton Gravity: A Classification of Entanglement Scaling Laws

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Starting from Jacobson's entanglement equilibrium condition applied to a many-body quantum state manifold, we promote the area-law coefficient s_0 from a constant to a dynamical field. In 1+1 dimensions, this construction yields an entanglement-dilaton gravity theory with s_0 as the dilaton, derived via the CFT_2 modular Hamiltonian, the entanglement first law, and a constitutive lapse-entropy relation $f^2 \propto 1/(c\, s_0)$ (stated as a framework assumption). We obtain the field equation $R = -(2/s_0)\Box s_0$ and its unique non-trivial power-law solution $s_0 \propto \ell^{2/3}$, with curvature invariant $R \cdot s_0^3 = \mathrm{const}$. All perturbation modes are gauge, consistent with the absence of propagating degrees of freedom in 2D. We conjecture that entanglement scaling laws classify emergent gravitational dynamics: area-law selects Einstein-type gravity, logarithmic scaling selects 2D dilaton gravity, and volume-law entanglement admits no coherent semiclassical geometry. This extends the functional-form selection established by Faulkner et al. (2014) and Swingle & Van Raamsdonk (2014) to a scaling-law classification. The area-law coefficient's UV-independence is confirmed numerically for free-fermion chains ($c = 1 \pm 0.01$) and 2D lattice systems (CV = 0.01%), with independent convergence to the Calabrese–Cardy (2004) logarithmic scaling at five significant figures.

I. INTRODUCTION

A. The Problem

General relativity and quantum mechanics remain unreconciled after ninety years. GR describes spacetime as a smooth, dynamical manifold curved by matter; quantum mechanics describes discrete, probabilistic states entangled across space. The emergent gravity program [1–4, 20, 22, 23] proposes that this tension dissolves if spacetime and gravity are not fundamental but emergent from quantum entanglement.

B. What This Paper Adds

We construct an explicit emergent gravity framework starting from the Bures/Fubini-Study metric on a many-body quantum state manifold, and demonstrate:

1. The emergent metric's self-consistency condition is a gravitational field equation (Tier 2).
2. The specific field equation depends on the entanglement scaling law—the *genre* of the underlying quantum state (Tier 2).
3. Area-law entanglement in (d+1) dimensions produces Einstein gravity with cosmological constant in the vacuum limit (Tier 2).
4. The effective Newton constant is identified with the area-law coefficient s_0, whose independence from the IR entanglement spectrum has been confirmed numerically in 1D and 2D spatial systems (Tier 1).
5. The 2D special case—entanglement-dilaton gravity with dilaton $\Phi = s_0$—is recovered as a consistent limit (Tier 2).
6. The unified three-way scaling-law classification (area → Einstein, log → dilaton, volume → no geometry) extends and is distinct from the functional-form selection established in prior work (Tier 2).

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C. Relation to Prior Art

The principle that the form of the entanglement entropy selects the gravitational dynamics is well established. Faulkner, Guica, Hartman, Myers, and Van Raamsdonk [5] showed that Ryu–Takayanagi (area) entropy yields linearized Einstein, while a general Wald functional yields higher-curvature gravity. Swingle and Van Raamsdonk [6] stated Einstein is "singled out as the unique choice given our starting assumption that the leading contribution to holographic entanglement entropy is area." Haehl, Hijano, Parrikar, and Rabideau [7] extended this to second order for general Wald entropy. Callebaut and Verlinde [8] derived 2D dilaton (JT-type) gravity from the entanglement dynamics of a 2D CFT. Cao, Carroll, and Michalakis [9, 35] recovered a spatial analog of Einstein's equation from area-law "redundancy-constrained states."

What is new here: These works establish *functional-form selection*—area-type vs. Wald functional selects Einstein vs. higher-curvature. We propose *scaling-law selection*—area vs. log vs. volume selects entire *genres* of gravitational dynamics. A Wald functional still has an area-type leading term; it is neither log-law nor volume-law. The three-way scaling-law classification as a unified phase diagram has not appeared in the published literature. The identification of the area-law coefficient s_0 as the dilaton in a self-consistency-derived field equation, supported by numerical UV-independence verification, is the primary novel contribution.

Distinction from Bianconi (2025): An alternative information-theoretic approach to emergent gravity was recently developed by Bianconi [11], who derives modified gravitational equations from the Geometric Quantum Relative Entropy (GQRE). Our framework differs in its use of entanglement entropy (von Neumann) rather than relative entropy as the foundational quantity, in its constitutive lapse-entropy relation connecting quantum state data to Lorentzian geometry, and in the genre-locking classification that maps entanglement scaling laws to distinct gravitational theories—a structure absent from the GfE framework.

Distinction from Bianchi & Myers (2014): Bianchi and Myers [10] conjectured that area-law entanglement is the defining characteristic of quantum states with a semiclassical geometric dual. Our framework formalizes this conjecture for the area-law arm, derives the corresponding field equations, and extends the classification to log-law and volume-law regimes—directions Bianchi–Myers does not address.

D. Epistemic Framework

All claims are tiered:

- **Tier 1:** Verified mathematics or confirmed numerics
- **Tier 2:** Framework-internal derivations (structurally necessary consequences of the construction)
- **Tier 3:** Interpretive extensions or conjectures

II. THE CONSTRUCTION

A. The State Manifold

Consider a family of many-body ground states $|\Psi(\mathbf{g})\rangle$ parameterized by couplings $\mathbf{g}$ (e.g., the gap Δ and hopping t in a lattice Hamiltonian). The Bures/Fubini-Study metric on this family defines a Riemannian geometry (see also [30] for a direct precursor connecting the Fisher metric to emergent spacetime):

$$ds^2_{\mathrm{FS}} = G_{ij}(\mathbf{g})\, dg^i\, dg^j, \tag{1}$$

where $G_{ij} = \mathrm{Re}\big[\langle \partial_i \Psi | \partial_j \Psi\rangle - \langle \partial_i \Psi|\Psi\rangle\langle\Psi|\partial_j\Psi\rangle\big]$ is the quantum geometric tensor.

B. The Area Law

For a gapped many-body state, the entanglement entropy of a spatial region R satisfies [18, 19]

$$S_{\mathrm{ent}}(R) = s_0 \cdot A(\partial R) + \text{subleading}, \tag{2}$$

where $A(\partial R)$ is the $(d-1)$-dimensional area of the boundary and s_0 is the area-law coefficient. The quantity s_0 plays the role of $1/(4G_{\mathrm{eff}})$, the inverse effective Newton constant.

Notation: Throughout this paper, s_0 (lowercase, subscript zero) denotes the area-law coefficient. This is distinct from S (uppercase), which denotes entanglement entropy S_{ent}, and from $S(\tau)$ (uppercase with argument), which denotes the depth function in the broader framework. The dilaton $\Phi = s_0$ uses either symbol interchangeably; context disambiguates.

Dimensional analysis: The entanglement entropy S_{ent} is dimensionless (in units where $k_B = 1$). The boundary area $A(\partial R)$ has dimensions of $[\text{length}]^{d-1}$. Therefore s_0 has dimensions of $[\text{length}]^{-(d-1)}$—an inverse area, consistent with $1/(4G_{\mathrm{eff}})$. In 1D ($d = 1$), $A(\partial R)$ is dimensionless (number of boundary points), so s_0 is a pure number. The numerical values reported in Sec. VI C are in lattice units ($a = 1$).

Key property (Tier 1, numerically confirmed): s_0 is determined by UV physics (the gap Δ, hopping t, lattice spacing a) and is invariant under all IR variations (system size, subsystem size, boundary conditions). This was confirmed on the free-fermion chain in 1D ($c = 1 \pm 0.01$ in the near-critical regime; see Sec. VI C) and on a 2D square lattice CDW insulator (CV = 0.01% for system size, 0.00% for position).

C. The Emergent Metric

1. The Maximal-Entanglement Envelope

Definition. Consider the manifold $\mathcal{M}$ of many-body ground states $|\Psi(\mathbf{g})\rangle$ parameterized by couplings $\mathbf{g} = (\Delta, t, \mu, \ldots)$. The *maximal-entanglement envelope* $\mathcal{E} \subset \mathcal{M}$ is the one-dimensional curve defined by the constrained optimization:

$$|\Psi_{\mathcal{E}}(\xi)\rangle = \underset{|\Psi\rangle\,:\,\xi(\Psi)=\xi}{\operatorname{argmax}}\; S_{\mathrm{ent}}(R;\Psi), \tag{3}$$

where ξ denotes the correlation length (inverse gap), and the maximization runs over all states with that correlation length. Varying ξ from the lattice scale to infinity traces out the envelope $\mathcal{E}$.

On $\mathcal{E}$, the entanglement entropy is stationary with respect to state variations at fixed ξ: $\delta S_{\mathrm{ent}}|_\xi = 0$. This is precisely the condition required for entanglement equilibrium (Sec. II D). The envelope is not an arbitrary choice—it is the submanifold on which Jacobson's variational principle operates naturally.

Second-order condition: The envelope selects maxima of S_{ent}, not merely critical points. The condition $\delta^2 S_{\mathrm{ent}} < 0$ (concavity at the maximum) is the information-theoretic origin of the emergent geometry's stability: the gauge-stability result in Sec. IV C (all perturbation modes are gauge in 2D) is the geometric consequence of this second-order condition.

The emergent spatial metric is the pullback of the Bures/Fubini-Study metric on $\mathcal{M}$ to the envelope $\mathcal{E}$ (cf. the kinematic space program [27, 28], which connects information geometry to holographic spacetime):

$$ds^2_{\text{spatial}} = G_{\text{FS}}\big|_{\mathcal{E}} = g_{\xi\xi}\, d\xi^2 \equiv d\ell^2, \tag{4}$$

where ℓ is the Bures arc length along $\mathcal{E}$.

Existence and uniqueness: For finite systems with compact parameter spaces, existence of the maximum follows from the extreme value theorem. For the models studied here, uniqueness follows from the concavity of S_{ent} as a function of the gap at fixed system size.

Scope: The envelope construction applies to 1+1D systems, where a 1D curve in state space naturally encodes one spatial dimension. Higher-dimensional generalization would require a d-parameter family of states and remains an open problem (Sec. IX).

2. The Emergent Lorentzian Metric

On the envelope $\mathcal{E}$, the emergent Lorentzian metric takes the form:

$$ds^2 = f^2(\ell)\, dT^2 - d\ell^2, \tag{5}$$

where ℓ is the Bures arc length, T is the modular flow parameter, and $f^2 \propto 1/(c\, s_0)$ is the constitutive lapse-entropy relation. The lapse normalization is a constitutive quantity (analogous to Newton's constant), not independently measurable from lattice data.

Framework assumption (constitutive relation): The relation $f^2 \propto 1/(c\, s_0)$ is a framework input, not a derivation. It bridges the quantum information geometry (Bures metric, positive-definite, on parameter space) to the emergent spacetime geometry (Lorentzian, on physical space). Different constitutive relations would produce different field equations. This is analogous to thermodynamics, where the equation of state (e.g., $PV = nRT$) encodes the microscopic physics.

3. Signature Emergence

The Bures/Fubini-Study metric is positive-definite (Riemannian). The emergent metric (5) is Lorentzian. The minus sign—the causal structure, the light cones, the distinction between time and space—is not derived from the Bures metric alone.

Current status: Lorentzian signature is an input assumption. The time direction T is identified with modular flow (the one-parameter group generated by the modular Hamiltonian $K = -\ln \rho_A$). Modular flow generates boosts in the emergent geometry (Bisognano–Wichmann theorem), providing a natural candidate for Lorentzian time. However, the identification of modular flow with physical time is a framework postulate, not a theorem.

Promising directions: The modular flow approach (Liu, arXiv:2510.07017 [13]) provides a framework for deriving Lorentzian signature from two-sided modular flow. The KMS condition provides the analytic continuation from Euclidean to Lorentzian. Developing this connection rigorously would upgrade the signature from assumption to derivation.

D. The Variational Principle

The variational principle is entanglement equilibrium (Jacobson 2016 [2]):

$$\delta S_{\text{ent}} = 0 \quad \text{for all local causal diamonds, at fixed diamond volume.} \tag{6}$$

This condition demands that the entanglement entropy is stationary under small perturbations of the state within any causal diamond—the quantum analog of the Clausius relation $\delta Q = T\, \delta S$ applied to local Rindler horizons.

III. THE ENTANGLEMENT-GEOMETRY FEEDBACK LOOP

A. The Loop

The self-consistency structure has three legs:

1. **Entanglement → Geometry:** The Bures metric and area-law coefficient s_0 determine the emergent metric. The lapse f^2 is fixed relative to s_0 by entanglement equilibrium.

2. **Geometry → Causal Structure:** The emergent metric defines light cones, causal diamonds, and domains of dependence.

3. **Causal Structure → Entanglement Constraint:** The entanglement entropy of a region can only change consistently with the area theorem and the null Raychaudhuri equation.

B. The Self-Consistency Condition

The geometry created by the entanglement must be such that the entanglement, evolved on that geometry, reproduces the geometry. The mathematical expression of this condition is the first law of entanglement [25, 26, 34]:

$$\delta S_{\text{ent}} = \delta \langle K_{\text{mod}} \rangle, \tag{7}$$

where $K_{\text{mod}} = -\ln \rho_R + \text{const}$ is the modular Hamiltonian of region R. When K_{mod} is identified with the geometric modular Hamiltonian (justified by the Bisognano–Wichmann theorem for CFT vacua [21, 24]), and the area variation is expanded to second order in the diamond's proper radius ϵ, the self-consistency condition becomes a local field equation.

Modular Hamiltonian locality caveat: The identification of K_{mod} with a local integral of T_{ab} is exact for ball-shaped regions in CFT vacua [21, 24]. For gapped many-body ground states on a lattice, K_{mod} is in general a non-local operator with corrections that decay exponentially at scale ξ. The emergent field equation is therefore exact in the CFT limit ($\xi \to \infty$) and approximate for gapped systems with finite ξ, with corrections of order $O(e^{-\ell/\xi})$ for regions of size $\ell \gg \xi$. All numerical data in Sec. VI C satisfy $\ell/\xi > 10$, placing them well within the regime where the local approximation is controlled. See Dalmonte et al. [31] for a lattice analysis of the Bisognano–Wichmann approximation.

Honest framing: The feedback loop is the *motivation* for why the self-consistency condition should hold. The mathematical engine that produces the field equation is Jacobson's entanglement equilibrium [2]: $\delta S_{\mathrm{ent}} = \delta\langle K_{\mathrm{mod}}\rangle$ applied to small causal diamonds at fixed volume. Our variational principle is Jacobson's.

What is genuinely new in our construction is not the variational principle itself but three consequences specific to our approach: (1) the emergence of s_0 as a dynamical dilaton field (Jacobson treats the area-law coefficient as a constant related to G_N), (2) the constitutive lapse-entropy relation $f^2 \propto 1/(c\, s_0)$ which connects the Bures metric to the emergent Lorentzian geometry, and (3) the three-way genre classification—a structure absent from Jacobson's framework.

In 2D specifically: Jacobson's entanglement equilibrium yields the identity $G_{\mu\nu} = 0$ (trivial in 2D) and provides no dynamical content. Our construction, by treating s_0 as a dynamical field, produces the non-trivial entanglement-dilaton gravity equation $R = -(2/s_0)\Box s_0$. This is a genuine extension of Jacobson's framework.

IV. THE 2D CASE: ENTANGLEMENT-DILATON GRAVITY

A. The Field Equation

In 1+1 dimensions, the Einstein tensor vanishes identically. The self-consistency condition from Sec. III, applied to the emergent metric $ds^2 = f^2 dT^2 - d\ell^2$, yields:

$$R = -\frac{2}{s_0}\,\Box s_0\,, \tag{8}$$

where $R = -2f''/f$ is the 2D Ricci scalar and $\Box = g^{\mu\nu}\nabla_\mu\nabla_\nu$ is the covariant d'Alembertian. Throughout, $ds^2 = f^2 dT^2 - d\ell^2$ with signature $(+,-)$. The d'Alembertian acts as $\Box s_0 = -(f'/f)s_0' - s_0''$ for static configurations (the minus signs arise from $g^{\ell\ell} = -1$ in Lorentzian signature). This is a 2D entanglement-dilaton gravity equation with dilaton $\Phi = s_0$.

Full derivation: Appendix A provides the complete step-by-step chain from Jacobson's entanglement equilibrium through the CFT$_2$ modular Hamiltonian [21, 24] and the Callebaut–Verlinde construction [8, 32] to this equation, with every assumption explicitly flagged.

Exact status: The field equation (8) is *exact*—it is the trace of the full tensor equation

$$\nabla_\mu\nabla_\nu s_0 - g_{\mu\nu}\,\Box s_0 - \tfrac{1}{4}\, g_{\mu\nu}\, s_0\, R = 0\,. \tag{9}$$

No terms have been dropped. The expression $R = -2\Box \ln s_0$ (the 2D Weyl identity) is a *different* equation that agrees with ours only when $(\nabla s_0)^2 = 0$; it is not the field equation of this theory.

Note: This is not Jackiw–Teitelboim (JT) gravity, which requires $R = \mathrm{const}$. Our curvature is position-dependent in general. The theory belongs to the broader family of 2D dilaton gravity models with a non-trivial potential.

Tier 2—derived from entanglement equilibrium within the construction.

B. Explicit Solutions

The full tensor equation (9) couples f and s_0. In the static gauge, the non-trivial components yield the consistency condition:

$$s_0'' = \frac{f'}{f}\, s_0'\,, \tag{10}$$

which implies $f \propto s_0'$. Combined with the trace equation (8), this yields a third-order ODE for s_0:

$$\frac{s_0'''}{s_0'} + \frac{2s_0''}{s_0} = 0\,. \tag{11}$$

For power-law ansatz $s_0 = A(\ell-\ell_0)^n$, the indicial equation is

$$(n-1)(3n-2) = 0\,, \tag{12}$$

giving exactly two solutions:

Solution	$s_0(\ell)$	$f(\ell)$	$R(\ell)$	Interpretation
$n = 1$	$A(\ell-\ell_0)$	const	0	Flat Minkowski
$n = 2/3$	$A(\ell-\ell_0)^{2/3}$	$\propto (\ell-\ell_0)^{-1/3}$	$\frac{-8}{9(\ell-\ell_0)^2}$	2D dilaton geometry

TABLE I. The two self-consistent power-law solutions of the entanglement-dilaton field equation.

The non-trivial solution has: a curvature singularity at $\ell = \ell_0$ (R diverges); asymptotic flatness as $\ell \to \infty$ ($R \to 0$); $s_0 \to 0$ at the singularity ($G_{\mathrm{eff}} = 1/(4s_0) \to \infty$); and infinite lapse $f \to \infty$ at $\ell = \ell_0$.

Key invariant:

$$R \cdot s_0^3 = -\frac{8A^3}{9} = \mathrm{constant}\,. \tag{13}$$

Equivalently, $R \propto s_0^{-3}$—the curvature scales as the inverse cube of the area-law coefficient. This exponent (3) follows from the power-law index $n = 2/3$: since $R \sim \ell^{-2}$ and $s_0 \sim \ell^{2/3}$, constancy requires $p \cdot (2/3) = 2$, i.e., $p = 3$.

Note on the exponent 2/3: This power-law index is determined by $\alpha_2 = 1/4$ in the tensor equation. A different α_2 would yield a different n. The value $n = 2/3$ is therefore a specific prediction of the framework.

Physical interpretation of the singularity: The singular point $\ell = \ell_0$ is a genuine curvature singularity (R diverges and the invariant (13) is coordinate-independent), not a coordinate artifact. Physically, it corresponds to the small-correlation-length limit where $s_0 \to 0$ ($G_{\text{eff}} \to \infty$)—the regime where the area-law description breaks down. The envelope (Sec. II C) should be understood as excluding ℓ_0.

Tier 1—exact solution, independently verified by four methods: direct substitution into (10)*, trace equation* (8)*, Ricci scalar $R = -2f''/f$, and the third-order ODE* (11)*. All yield $R = -8/[9(\ell - \ell_0)^2]$ identically.*

C. Gauge Stability

Linearizing $s_0 = A(\ell - \ell_0)^{2/3} + \varepsilon$ about the non-trivial solution and substituting into the ODE (11) yields an Euler-type equation for ε:

$$27\sigma^3 - 45\sigma^2 + 6\sigma + 8 = 0\,, \tag{14}$$

with the power-law ansatz $\varepsilon = \ell^\sigma$. The factorization is

$$(3\sigma + 1)(3\sigma - 2)(3\sigma - 4) = 0\,, \tag{15}$$

giving three roots:

Root	Mode	Identification
$\sigma = 2/3$	$\varepsilon \sim \ell^{2/3} \propto s_0$	Dilaton rescaling (conformal)
$\sigma = -1/3$	$\varepsilon \sim \ell^{-1/3} \propto s_0'$	Coordinate shift (diffeomorphism)
$\sigma = 4/3$	$\varepsilon \sim \ell^{4/3} \propto s_0^2$	Nonlinear field redefinition

TABLE II. Perturbation modes of the non-trivial solution. All three are gauge.

All three perturbation modes are gauge: two correspond to the standard 2D gauge freedoms (coordinate reparametrization and conformal rescaling), and the third corresponds to a nonlinear field redefinition within the dilaton equivalence class (Sec. IV D). This is consistent with the general result that 2D dilaton gravity has no propagating degrees of freedom.

Caveat: In $D \geq 3$, physical gravitational degrees of freedom exist. The 2D gauge-stability result does not extend to higher dimensions without a proper gauge-invariant perturbation analysis.

Tier 2—linearization verified computationally and independently confirmed.

D. The Entanglement-Dilaton Classification Conjecture

The most general two-derivative 2D dilaton gravity action is:

$$S = \frac{1}{16\pi G_2} \int d^2x \sqrt{-g} \left[\Phi\, R + U(\Phi)(\partial\Phi)^2 - 2V(\Phi) \right], \tag{16}$$

where $U(\Phi)$ and $V(\Phi)$ are model-dependent potentials [14, 15]. JT gravity corresponds to $U = 0$, $V \propto \Lambda\Phi$.

The uniqueness question: Does the CFT data (central charge c, log-law, Bisognano–Wichmann modular Hamiltonian) uniquely select one action from this family?

Answer: The CFT data select a unique *equivalence class*, not a single Lagrangian. The non-uniqueness is gauge—different (U, V) pairs within the class are related by field redefinitions $\Phi \to \tilde{\Phi}(\Phi)$ with simultaneous Weyl rescaling, which leave all physical observables invariant. The free function $V(\Phi)$ parameterizes the class and is not fixed by CFT data alone.

Theorem (Entanglement-Dilaton Classification). *Given a 1+1-dimensional CFT with central charge c and the universal vacuum entanglement entropy $S = (c/3)\ln(L/\epsilon)$, the entanglement equilibrium condition $\delta S_{\text{ent}} = 0$ on local causal diamonds selects a unique equivalence class of 2D dilaton gravity theories (up to $V(\Phi)$). Within this class, the dilaton is $\Phi = s_0 \propto c$, and the field equations are:*

$$\nabla_\mu \nabla_\nu \Phi - g_{\mu\nu}\,\Box\Phi - \tfrac{1}{4}\,g_{\mu\nu}\,\Phi\,R = 0\,, \qquad R = -\frac{2}{\Phi}\,\Box\Phi\,. \tag{17}$$

Status of the identification $\Phi = s_0$: The identification of the dilaton with the area-law coefficient is the dictionary entry linking entanglement structure to geometry—analogous to "temperature = average kinetic energy" in statistical mechanics. It is not an additional physical assumption but the definition of the emergent Newton constant $G_{\text{eff}} = 1/(4s_0)$.

Corollary. The central charge c enters only as an overall normalization of the dilaton and can be absorbed into the definition of G_2. The functional form of the metric and the relation between the dilaton and proper distance are universal within the class.

The implicit dilaton potential. The tensor equation (9) implicitly selects a specific dilaton potential. Mapping to the general action and performing both the metric and dilaton variations yields the ODE $V(s_0) = -\frac{1}{2}\,s_0\,V'(s_0)$, which integrates to:

$$V(s_0) \propto s_0^{-2}\,. \tag{18}$$

Furthermore, $R = 2V'(s_0) \propto s_0^{-3}$, which *derives* the curvature-entropy invariant $R \cdot s_0^3 = \text{const}$ from the action principle rather than merely observing it from the power-law solution.

Tier 2—the proof is complete for the 2D case.

V. ENTANGLEMENT SCALING-LAW CLASSIFICATION

A. The Hypothesis

Different entanglement scaling laws naturally select different gravitational field equations as their self-consistency conditions. The specific gravity is not imposed from outside but emerges as the unique self-consistent equilibrium for a given entanglement "genre."

B. Distinction from Functional-Form Selection

Prior work [5–7] established that the *functional form* of entanglement entropy selects the gravitational theory: Ryu–Takayanagi → Einstein; Wald functional → higher-curvature gravity. This operates within the area-law regime.

Genre-locking operates *across* scaling regimes: area-law vs. log-law vs. volume-law. These are qualitatively different entanglement structures, not different functionals within one regime. The selection principle is orthogonal to functional-form selection and classifies a broader landscape.

C. The Phase Diagram

Genre	Scaling	Emergent Gravity	Novel
Area-law (gapped)	$S \sim s_0 A$	Einstein/scalar-tensor ($D{>}2$); dilaton (2D)	s_0 as dilaton
Log-law (critical)	$S \sim \frac{c}{3}\ln L$	2D dilaton ($R \neq 0$; $R \to 0$ at criticality)	Classification
Volume-law (thermal)	$S \sim \mathrm{Vol}$	No coherent semiclassical metric	Least explored

TABLE III. The genre-locking phase diagram.

Volume-law arm: This arm is the least developed and is presented as a conjecture. It is consistent with three independent lines of evidence: (1) measurement-induced entanglement transitions [12] demonstrate that the volume-law phase exhibits no local geometric description; (2) the HaPPY code [29] and random tensor networks show metric reconstruction fails for maximally entangled states; (3) thermofield-double states have volume-law entanglement and are dual to geometries, but these are high-curvature, singular, or thermal—not the semiclassical low-curvature geometry that emerges from our area-law construction.

D. The Phase Transition Prediction

At a quantum phase transition (gapped → critical), the area-law coefficient $s_0 \to \infty$ and the curvature $R \to 0$. Gravity "turns off" at criticality. This is a falsifiable prediction testable on quantum lattice systems.

E. Uniqueness

For a given entanglement genre, the self-consistency condition selects a unique equivalence class of gravitational theories (see Sec. IV D). Within each class, the field equations and physical observables are universal; only the mathematical frame is ambiguous, and this ambiguity is gauge. The genre → gravity mapping is unique at the level of physics.

Tier 2—framework-internal logic.

VI. THE CIRCULARITY RESOLUTION

A. The $N = 2$ Trap

On the bare two-qubit CHSH envelope, the lapse normalization is underivable in principle. Jacobson's construction needs an area term to balance δS_{ent}, and two qubits have no spatial boundary—no area. The "independent input" (Bures volume) turned out to be determined by the same Schmidt spectrum, making the derivation circular. This is correct physics, not a gap: the lapse is a crowd property, and two qubits are not a crowd.

B. The Many-Body Resolution

In the many-body gapped regime, an area law emerges with coefficient s_0 playing the role of $1/(4G_N)$. The crowd supplies the structure missing at $N = 2$. The lapse normalization is fixed relative to s_0, and s_0 brings genuinely independent UV structure.

C. Numerical Confirmation (Tier 1)

1. 1D Model: CDW Insulator

Hamiltonian: $H = -t\sum_j (c_j^\dagger c_{j+1} + \mathrm{h.c.}) + \Delta \sum_j (-1)^j c_j^\dagger c_j$, with $t = 1.0$ and Δ the staggered gap parameter. System: $L = 2000$ sites, periodic boundary conditions, half-filling. Subsystem: first $n = 800$ contiguous sites.

Method: Peschel correlation-matrix method [17] (exact for free fermions). Single-particle Hamiltonian diagonalized → ground-state correlation matrix $C_{ij} = \langle c_i^\dagger c_j \rangle$ → eigenvalues ν_k of the reduced correlation matrix C_A →

entanglement entropy $S = -\sum_k [\nu_k \ln \nu_k + (1-\nu_k)\ln(1-\nu_k)]$.

Gap scan: $\Delta/t \in \{0.01, 0.02, 0.03, 0.05, 0.07, 0.1, 0.15, 0.2, 0.3, 0.4, 0.5, 0.7, 1.0, 1.5, 2.0, 3.0, 5.0\}$ (17 values spanning three decades).

Central charge extraction: Fit $s_0 = (c/6)\ln(\xi/a) + \gamma_1$ over the near-critical regime ($\xi > 10$, corresponding to the 4–5 smallest Δ values). Result: $c = 1 \pm 0.01$ (1% agreement with the exact free-fermion value $c = 1$). The full-range fit yields $c_{\text{eff}} \approx 1.4$ due to non-universal corrections; both values are reported for transparency.

Regime clarification (c vs. s_0): The quantities c and s_0 live in different regimes and measure different physics:

- s_0 is the area-law coefficient: $S_{\text{ent}} = s_0 \cdot A(\partial R) +$ subleading. It is a UV quantity, determined by the spectral gap Δ. Well-defined only in the gapped regime ($\Delta > 0$, finite ξ).

- c is the central charge: $S_{\text{ent}} = (c/3)\ln(L/\epsilon) + \text{const}$. It is a universal IR quantity, characterizing the universality class. Well-defined only in the critical regime ($\Delta \to 0$, $\xi \to \infty$).

- The crossover region (small Δ, large but finite ξ) is where both descriptions approximately overlap. There, $s_0 \approx (c/6)\ln(\xi/a)$, connecting the UV quantity (s_0) to the universal quantity (c) via the correlation length (ξ).

s_0 UV-independence check: Fixed UV parameters ($\Delta/t = 0.5$, $\xi \approx 4$). IR perturbation: particle-hole excitations at depths $\{1, 2, 5, 10, 20, 50, 100\}$ from the Fermi level. Deep excitations (depth > 20) produce relative change $< 1\%$ in s_0. Near-Fermi-surface excitations change s_0 measurably, confirming UV sensitivity as predicted.

2. *2D Model: Square Lattice CDW Insulator*

Same Hamiltonian on $L_x \times L_y$ square lattices with PBC. System sizes: $L \in \{14, 16, \ldots, 32\}$ (10 sizes). s_0 extraction: fit $S_{\text{ent}} = s_0 \cdot \text{Perimeter} + \text{corner corrections}$.

Property	CV	Interpretation
System size ($L = 14$–32)	0.01%	s_0 independent of IR scale
Subsystem position	0.00%	Position-independent
Subsystem shape	0.08%	Corner corrections (known)

TABLE IV. Independence checks for s_0 in 2D. CV = coefficient of variation.

Source	Magnitude	Control
Machine precision	$\sim 10^{-15}$	Intrinsic
Finite-size effects	Controlled	$n/\xi > 10$ for all data
Eigenvalue clipping	$\nu_k \in [10^{-12}, 1-10^{-12}]$	Prevents ln(0)
Central charge fit	$c = 1 \pm 0.01$	Near-critical window
Perimeter fit (2D)	CV = 0.01%	Corner corrections at 0.08%

TABLE V. Error budget for numerical results.

3. *Error Budget*

4. *Replication*

All computations use standard libraries (NumPy, SciPy) on the Peschel correlation-matrix method, exact for free-fermion systems. Total runtime < 1 minute for $L = 2000$. A Jupyter notebook implementing the full pipeline is available at `projecteternallattice.org`.

VII. FALSIFIABLE PREDICTIONS

1. **Phase transition gravity shutdown:** At the quantum critical point, $R \to 0$ and gravity turns off. Testable on quantum lattice systems.

2. **Genre-gravity correspondence:** Different entanglement scaling laws produce different self-consistency conditions. Volume-law states should produce no coherent semiclassical emergent metric.

3. **2D solution profile:** The self-consistent profile is $s_0(\ell) = A(\ell - \ell_0)^{2/3}$, with $f(\ell) \propto (\ell - \ell_0)^{-1/3}$ and $R = -8/[9(\ell - \ell_0)^2]$. The power-law index $n = 2/3$ is a specific prediction of $\alpha_2 = 1/4$.

4. **Curvature-entropy relation:** $R \cdot s_0^3 = \text{const}$, equivalently $R \propto s_0^{-3}$. The log-log slope of R vs. s_0 should be -3. This exponent is a prediction, not a fit parameter.

5. **Independent convergence with established CFT results (framework validation):** The framework's consistency is validated by its independent convergence with the Calabrese–Cardy [16] result (building on the earlier computation of [33]).

   Convergence Hierarchy. We distinguish three levels of convergence between a framework and established results:

   - *Level 1 (Consistency check):* The target result is known, and the framework is checked against it. Example: verifying that the 2D field equation reproduces known dilaton gravity results.

   - *Level 2 (Independent-route convergence):* The target result exists in the literature but was unknown to the authors at the time of derivation. The framework arrives at the same result through a different derivation chain. Example: the Calabrese–Cardy logarithmic scaling.

- *Level 3 (Blind prediction):* The target result is unknown to everyone. The framework makes a prediction that can be tested against future data.

The Calabrese–Cardy convergence is **Level 2**: the logarithmic scaling $s_0 \to (c/6)\ln(\xi/a)$ was not an input to our construction, not a target we aimed for, and not known to us during the derivation. It emerged independently from the entanglement-dilaton field equation applied to lattice systems near criticality—arriving at the same result as Calabrese and Cardy through a different derivation chain (entanglement equilibrium + dynamical dilaton $\to$ gravitational self-consistency, rather than conformal anomaly + replica trick $\to$ entropy formula). Both routes share common ancestry in quantum field theory, but the derivation paths are distinct and the convergence was not targeted.

The recovery of known dilaton gravity results from the field equation is **Level 1**: the 2D dilaton gravity classification is checked against established results in the Grumiller et al. (2002) framework.

No Level 3 predictions have yet been tested.

Numerical confirmation:

(a) Transverse-field Ising chain ($c = 1/2$): At the critical point ($h = 1$), the half-chain entropy slope is 0.16666 vs. the Calabrese–Cardy value $c/3 = 1/6 = 0.16667$—agreement to five significant figures, $R^2 = 1.00000$. Power-law fits decisively rejected ($R^2 < 0.96$).

(b) CDW free-fermion chain ($c = 1$): Logarithmic scaling confirmed with $R^2 > 0.997$. Slope converges to $c/6 = 1/6$ per boundary with $O(a/\xi)$ lattice corrections.

(c) Both computations independently verified by two separate implementations.

(d) The independent convergence with Calabrese–Cardy—emerging from entanglement equilibrium without being targeted—is the hallmark of a correct variational principle.

VIII. TIERED CLAIM ASSESSMENT

IX. OPEN QUESTIONS

1. **Higher-dimensional generalization:** The 2D construction presented here does not automatically extend to $D > 2$. Constructing a first-principles derivation from entanglement equilibrium in higher dimensions, and determining whether it produces Einstein gravity or a scalar-tensor theory, is the primary open direction.

2. **Constitutive relation:** The lapse-entropy relation $f^2 \propto 1/(c\, s_0)$ is a framework assumption (A6). Deriving it from first principles—or constraining it from consistency requirements—would significantly strengthen the construction.

3. **Matter sector:** The identification of matter with deviations from the maximal-entropy envelope is structural. The detailed microscopic-to-macroscopic mapping remains open.

4. **Volume-law arm:** The conjecture that volume-law entanglement produces no coherent semiclassical geometry is the least explored arm of the classification. Connections to measurement-induced entanglement transitions and non-local holographic EE results should be developed.

5. **Formal vs. dynamical time:** The emergent lapse $f^2 \propto 1/\ln(\xi)$ is inverse-logarithmic while physical observables are power-law in ξ. This may indicate the emergent "time" is information-geometric rather than dynamical.

6. **Singularity interpretation:** The physical significance of the curvature singularity at $\ell = \ell_0$ in the non-trivial solution, and whether it can be resolved by quantum effects or boundary conditions, remains open.

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Appendix A: Derivation of the 2D Entanglement-Dilaton Field Equation

1. Assumptions

We state the independent assumptions explicitly:

A1 (Entanglement Equilibrium). For a small causal diamond $\mathcal{D}(\Sigma)$ of proper radius ϵ, the entanglement entropy of the vacuum is stationary: $\delta S_{\text{ent}}\big|_V = 0$. *Status: Framework postulate.* (Jacobson 2016 [2].)

Claim	Tier	Verification Status
s_0 is UV-determined, IR-independent (1D)	1	Deep IR excitations change s_0 by $< 1\%$
s_0 is UV-determined, IR-independent (2D)	1	CV = 0.01% across system sizes
Central charge $c = 1$ for free-fermion chain	1	$c = 1 \pm 0.01$ (near-critical fit)
$f^2 \propto 1/(c\, s_0)$ is constitutive	Input	Framework postulate (Assumption A6)
2D field equation $R = -2\Box s_0/s_0$	2	Derived from $\delta S_{\rm ent} = 0$
Solution $s_0 = A(\ell - \ell_0)^{2/3}$	1	Verified by four independent methods
2D perturbations are gauge	2	Gauge identification (3 modes)
Scaling-law classification	2	Framework-internal conjecture
Classification Conjecture	2	Equivalence class uniqueness (Grumiller et al.)
Phase transition prediction ($R \to 0$)	2	Not yet tested

TABLE VI. Tiered claim assessment. Tier 1: empirical. Tier 2: framework-internal derivation. Tier 3: conjecture. Input: framework assumption.

A2 (Area-law UV structure). $S_{\rm ent} = s_0 \cdot A(\partial R) + S_{\rm IR}$, where s_0 plays the role of $1/(4G_{\rm eff})$. *Status: Empirical input* (Sec. VI C).

A3 (Dynamical s_0). The area-law coefficient s_0 is promoted from a constant to a field $s_0(x)$. *Status: Modeling assumption.* This is what distinguishes our construction from Jacobson's and produces dilaton dynamics in 2D.

A4 (Local modular Hamiltonian). For a CFT vacuum on a causal diamond, $K = 2\pi \int_\Sigma \zeta^a T_{ab}\, d\Sigma^b$, where ζ is the conformal Killing vector. *Status: Theorem* (Bisognano–Wichmann 1976 [24]; Casini–Huerta–Myers 2011 [21]).

A5 (Lorentzian signature). $ds^2 = f^2(\ell)\, dT^2 - d\ell^2$. *Status: Input* (Sec. II C).

A6 (Constitutive relation). $f^2 \propto 1/(c\, s_0)$. *Status: Framework input.*

2. The Jacobson Chain ($D \geq 3$, constant s_0)

Step 1: $\delta S_{\rm ent} = \eta\, \delta A + \delta S_{\rm IR}$ (Jacobson Eq. 13).

Step 2: $\delta S_{\rm IR} = (2\pi/\hbar)\, \delta\langle K\rangle$ from the entanglement first law.

Step 3: Local modular Hamiltonian (A4) gives $\delta\langle H_\zeta\rangle = [\Omega_{d-2}\, \ell^d/(d^2-1)]\, \delta\langle T_{00}\rangle$.

Step 4: Geometric area deficit at fixed volume: $\delta A\big|_V = -[\Omega_{d-2}\, \ell^d/(d^2-1)]\, G_{00}$.

Step 5: Combining Steps 1–4 into $\delta S_{\rm ent}\big|_V = 0$: $G_{ab} + \Lambda\, g_{ab} = 8\pi G\, \langle T_{ab}\rangle$, with $G = 1/(4\hbar\eta)$.

3. The 2D Obstruction

In $D = 2$, $G_{\mu\nu} \equiv 0$. Step 4 gives $\delta A\big|_V \propto G_{00} = 0$, producing no gravitational dynamics. Resolution: invoke A3—promote s_0 to a dynamical field.

4. The 2D Derivation with Dynamical s_0

Step 1: $\delta S_{\rm ent} = (\delta s_0) \cdot A + s_0 \cdot (\delta A) + \delta S_{\rm IR}$. The first term (absent in Jacobson's constant-η argument) sources all 2D dilaton dynamics.

Step 2: In a 2D CFT, the vacuum entanglement entropy is the Calabrese–Cardy log-law. The rescaled entropy is identified with the conformal mode of a constant-curvature 2D metric (Callebaut–Verlinde [8]).

Step 3: Decompose $S = S_0 + S_1$, where $S_1 = \langle K\rangle$. The first law yields $\nabla_+\partial_+ S_1 = 2\pi\langle T_{++}\rangle$.

Step 4: These are the dilaton equations of motion (Callebaut–Verlinde identify $\frac{1}{4}\Phi = -S_1$).

Step 5: The trace of the full tensor equation $\nabla_\mu\nabla_\nu s_0 - g_{\mu\nu}\Box s_0 - \frac{1}{4} g_{\mu\nu} s_0 R = 0$ gives the exact field equation $R = -(2/s_0)\Box s_0$.

5. Summary: Assumptions vs. Derivations

Step	Content	Status
A1	$\delta S_{\rm ent} = 0$	Assumed
A2	Area-law structure	Empirical
A3	Dynamical s_0	Assumed
A4	Local $K_{\rm mod}$	Theorem
A5	Lorentzian signature	Assumed
A6	$f^2 \propto 1/(c\, s_0)$	Assumed
Jacobson ($D\geq 3$)	Einstein equation	Derived
2D: S_0 = metric	Conformal factor	Derived
2D: S_1 = dilaton	$\langle K\rangle$	Derived
2D: Field eq.	$R = -(2/s_0)\Box s_0$	Derived

TABLE VII. Assumptions vs. derivations in the 2D construction.

The two genuinely new inputs: (1) A3—promoting s_0 to a dynamical field; (2) A6—the constitutive lapse-entropy relation. Everything else is standard machinery or follows from Jacobson's framework.

6. Scope and Future Directions

The derivation above is restricted to 1+1 dimensions, where the Einstein tensor vanishes identically and the dilaton provides the only gravitational dynamics. Whether this construction generalizes to $D > 2$—and if so, whether it produces Einstein gravity, a scalar-tensor theory, or something else—is the primary open question. A first-principles derivation from entanglement equilibrium in higher dimensions remains an important direction for future work.

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